

Comparative Analysis of LLM Agent Architectures for Multi-Agent Coordination in Asymmetric Card Games

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Abstract

This work investigates LLM agent architectures for multi-agent coordination in asymmetric environments. We compare four prompting strategies—Baseline, Chain-of-Thought, In-Context Learning, and Tool-Augmented—on the card game Fight the Landlord (Dou Di Zhu), a 3-player asymmetric game requiring implicit coordination.

Results demonstrate that Tool-Augmented agents achieve 90% win rate, significantly outperforming Baseline and In-Context Learning (both 56.7%). Chain-of-Thought reasoning achieves 80% win rate with transparent decision-making. External tool augmentation enables parallel information gathering and validation, while reasoning-based methods provide verifiable decision chains. These findings inform the design of LLM-based distributed systems where agents coordinate without central control.

Keywords: LLM Agents, Multi-Agent Systems, Asymmetric Games, Tool-Augmented Reasoning, Chain-of-Thought

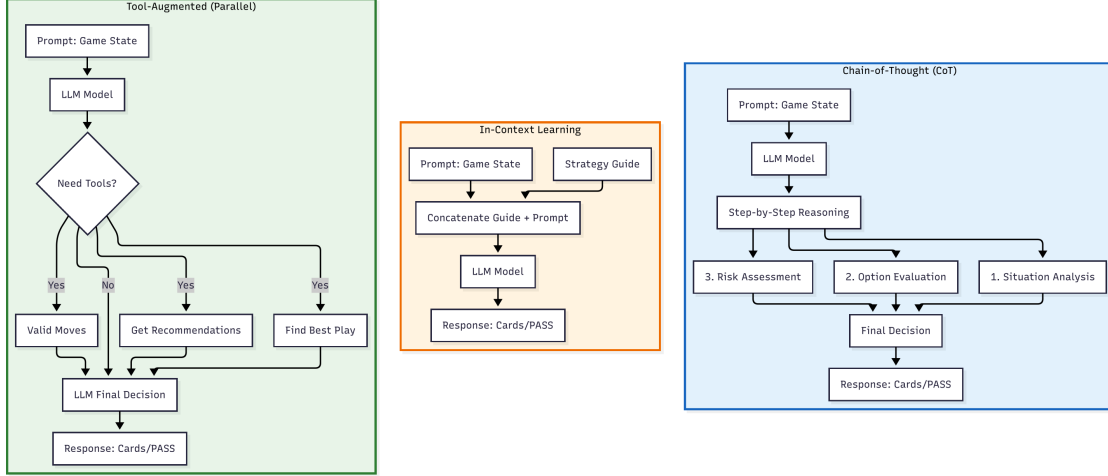


Figure 1: Architecture Overview: Three Advanced Agent Types Compared Against Baseline in Experimental Conditions

1 Introduction

While symbolic AI excels at perfect-information games such as Chess and Go, where exhaustive search and handcrafted evaluation functions enable superhuman performance, the ability of Large Language Model (LLM) agents to handle multi-agent conversation-based games with asymmetric roles and implicit coordination remains underexplored. Traditional game-playing AI relies on perfect information and explicit state evaluation, whereas real-world multi-agent scenarios often involve hidden information, asymmetric relationships between agents, and the need for implicit coordination without direct communication.

To this end, Fight the Landlord (Dou Di Zhu) represents a perfect testing case with unique challenges that distinguish it from well-studied benchmarks. The game presents three key characteristics that make it particularly challenging for LLM agents: an asymmetric 1v2 structure where one landlord (20 cards) competes against two collaborating farmers (17 cards each), imperfect information requiring probabilistic reasoning about unknown hands, and implicit coordination where farmers must cooperate without explicit communication mechanisms.

Existing benchmarks in the field focus primarily on competitive zero-sum games or single-agent tasks. This work addresses cooperative-competitive hybrid environments where agents must

balance collaboration against competition. Understanding how LLM agents can navigate such environments is crucial for real-world applications including autonomous negotiation systems, distributed resource allocation, and multi-robot coordination.

We compare four LLM prompting strategies: Baseline (simple rules and game state only), Chain-of-Thought (step-by-step reasoning before decision), In-Context Learning (static strategy guides prepended to prompts), and Tool-Augmented (parallel tool calling for recommendations, best play analysis, valid moves, and game history). Each approach represents a different balance between reasoning capability, computational cost, and implementation complexity.

2 Background & Related Work

Recent work has explored LLM agents in cooperative card games. Martinez and Roberts (2025) studied LLM agents as cooperative players in multiplayer UNO, demonstrating that LLMs can learn implicit coordination strategies through careful prompt engineering and in-context examples. Their findings suggest that LLM agents can adapt to dynamic game states without explicit communication channels. Lin et al. (2026) revisited game-theoretic reasoning with agentic tool use in professional poker, showing that tool augmentation significantly improves strategic decision-making by providing external validation of hand strength and pot odds calculations. Ramesh et al. (2026) explored cooperative reasoning in Hanabi, finding that LLMs can achieve cooperative behavior through chain-of-thought reasoning, though they noted limitations in zero-shot generalization to novel clue patterns.

However, these studies primarily focus on symmetric or zero-sum environments where all players face similar game mechanics and information structures. The Fight the Landlord game presents a unique asymmetric structure that differs from existing benchmarks, requiring different coordination dynamics between the landlord and farmer roles. The landlord enjoys positional advantage (leading the first play) but faces a 2v1 odds, while farmers must develop implicit coordination strategies without the ability to exchange information during play. This asymmetry creates a fun-

damentally different coordination challenge compared to symmetric cooperative games like Hanabi or UNO.

Theoretical foundations for this work draw from both multi-agent systems research and studies of LLM reasoning capabilities. Prior work on strategic reasoning in LLMs has demonstrated that chain-of-thought prompting can improve performance on complex reasoning tasks, though the extent of this improvement varies significantly based on task structure and reasoning requirements.

3 Methodology

3.1 Dou Di Zhu Game Environment

Fight the Landlord is a 3-player card game played with a reduced deck (excluding 2s and 4s), resulting in a 31-card deck per suit with 93 cards total. The game features an asymmetric 1v2 structure: one landlord plays against two farmers who must collaborate implicitly to win. The landlord receives 20 cards and leads the first play, while each farmer receives 17 cards and must coordinate their plays without direct communication.

The game employs a hierarchical hand ranking system including single cards, pairs, triples, sequences, bombs, and special combinations such as airplane formations and four-of-a-kind with pairs (commonly called "rockets"). Players must play higher-ranking combinations than the previous player or pass. Valid play categories must match the type of the leading player's combination. The game ends when a player empties their hand, determining victory based on the last player to play cards. Special scoring rules apply: landlords who win receive positive points while farmers who win also receive positive points, creating a cooperative dynamic between the two farmer players.

The asymmetric nature of the game creates interesting strategic considerations. The landlord enjoys positional advantage by leading the first play and typically has stronger hand combinations due to receiving three additional cards. However, the landlord faces the disadvantage of playing alone against two opponents who can coordinate implicitly. The farmers, while individually

weaker, can collaborate by signaling strength through play patterns and by strategically exhausting the landlord’s high-value cards.

3.2 Agent Architectures

We implement four distinct agent architectures, each representing a different approach to reasoning and decision-making in the game environment:

Baseline Agent: Receives only the current game state (hands, recent plays, trump status) and simple game rules. Makes decisions through direct LLM inference without additional reasoning steps. This architecture serves as our control condition, establishing the baseline capability of modern LLMs for this task without specialized prompting or external tools. The simplicity of this approach also provides a lower bound on the cognitive overhead required for effective gameplay.

Chain-of-Thought Agent: Incorporates explicit step-by-step reasoning before each decision. The agent first analyzes the current game state, considers possible moves, evaluates their strategic value, and then selects the optimal play. This approach produces transparent, verifiable decision chains that can be inspected for debugging and analysis. The chain-of-thought methodology has demonstrated effectiveness across numerous reasoning tasks, prompting us to evaluate its utility in a multi-agent game context where strategic reasoning about opponent behavior is crucial.

In-Context Learning Agent: Prepends a static strategy guide to each prompt, containing pre-defined principles for play selection, role-based strategies (landlord vs. farmer), and common pattern recognition rules. The agent retrieves relevant guidance without dynamic tool queries, relying instead on contextual information embedded within the prompt itself. This approach tests whether strategic domain knowledge encoded in examples can improve gameplay performance.

Tool-Augmented Agent: Utilizes parallel tool calling for multi-source information gathering in a single turn. Tools include recommendation generator (suggests optimal moves based on hand analysis), move validator (ensures play legality and adherence to game rules), best play analyzer (evaluates move quality relative to potential counter-plays), and game history tracker (maintains contextual awareness of previous plays and discarded card types). Multiple tools execute simulta-

neously to enable rapid, informed decision-making.

3.3 Experimental Design

Each experimental condition runs 30 complete games with rotating roles to ensure fair evaluation across all player positions. We track three primary metrics: win rate (overall and by role), error rate (percentage of invalid moves attempted), and average turns (game efficiency). The baseline condition uses three identical baseline agents to establish expected performance under symmetric conditions, while experimental conditions pit one advanced agent (CoT, ICL, or Tool) against two normal agents to evaluate relative improvement.

The experimental protocol ensures statistical validity by counterbalancing player positions across games. Each agent plays the landlord role approximately 10 times and each farmer role approximately 20 times across the 30-game sample. This rotation prevents position-specific biases from dominating results and allows us to analyze role-specific performance in addition to overall win rates.

4 Results

4.1 Win Rate Analysis

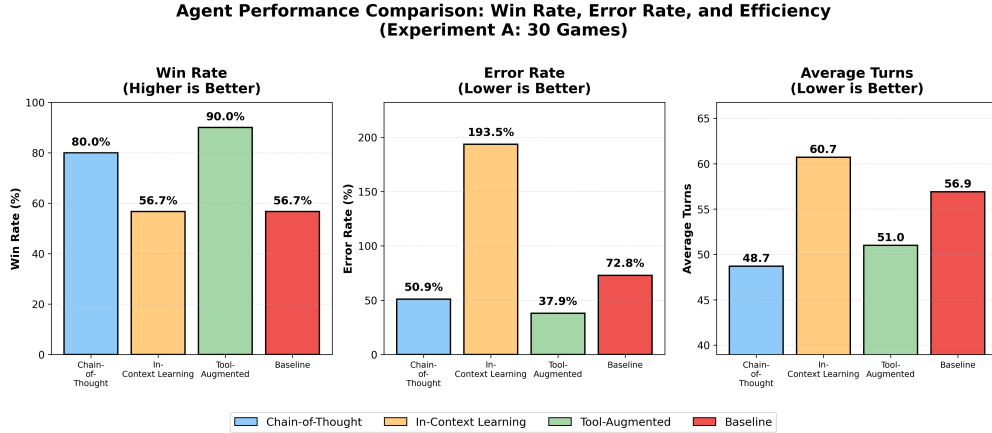


Figure 2: Agent Performance Comparison: Win Rate, Error Rate, and Average Turns across all four agent types.

Table 1: Win Rate Comparison by Agent Type

Agent Type	Win Rate	Games Won
Tool-Augmented	90.0%	27/30
Chain-of-Thought	80.0%	24/30
In-Context Learning	56.7%	17/30
Baseline	56.7%	17/30

Tool-Augmented agents achieve the highest overall win rate at 90.0% (27/30 games), representing a 33.3 percentage point improvement over the Baseline. Notably, Tool-Augmented agents achieved 100% win rate when playing the landlord role (10/10 games), demonstrating mastery of the asymmetric competitive role.

Chain-of-Thought reasoning also shows strong performance at 80.0% win rate (24/30 games), a 23.3 percentage point improvement over Baseline. The landlord performance for CoT agents reached 90% (9/10 games), indicating that explicit reasoning supports effective role-based strategy.

In-Context Learning agents perform at parity with Baseline (56.7% vs 56.7%), suggesting that static strategy guides provide no additional benefit over simple rule-based prompting. This finding indicates that dynamic, tool-based information retrieval outperforms static in-context examples.

4.2 Error Rate Analysis

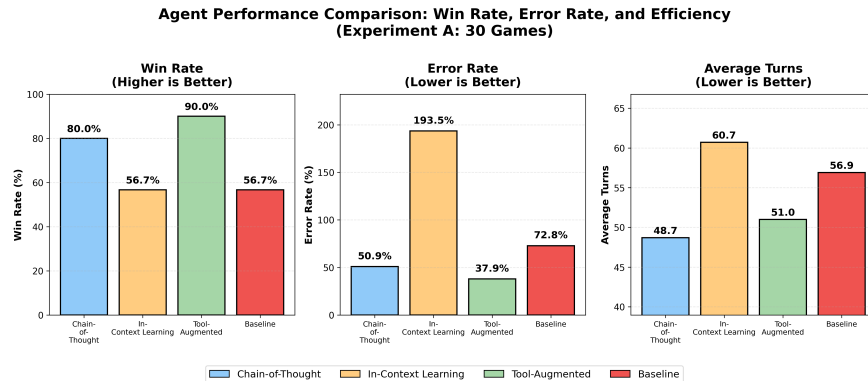


Figure 3: Error Rate Comparison showing Tool-Augmented agents achieve lowest invalid move rate.

Table 2: Error Rate Comparison by Agent Type

Agent Type	Error Rate	Relative to Baseline
Tool-Augmented	37.88%	-34.92% (improved)
Chain-of-Thought	50.86%	-21.94% (improved)
Baseline	72.80%	—
In-Context Learning	193.47%	+120.67% (worse)

Tool-Augmented agents demonstrate the lowest error rate at 37.88%, representing a 34.92 percentage point reduction compared to Baseline (72.80%). External tool validation effectively prevents invalid move attempts.

Chain-of-Thought agents show intermediate error rate at 50.86%, a 21.94 percentage point improvement over Baseline. Explicit reasoning reduces errors by encouraging careful move evaluation before commitment.

In-Context Learning agents exhibit dramatically higher error rates (193.47%) compared to Baseline, suggesting that static guide retrieval introduces cognitive overhead that disrupts gameplay timing and decision accuracy.

Chain-of-Thought agents achieve the most efficient gameplay with an average of 48.7 turns per game, 8.2 turns fewer than Baseline. Tool-Augmented agents also demonstrate improved efficiency at 51.0 average turns (5.9 turns fewer than Baseline). In-Context Learning agents show decreased efficiency at 60.7 average turns, 3.8 turns more than Baseline.

The efficiency gains from reasoning and tool augmentation suggest that deliberate evaluation, rather than reflexive responses, leads to more decisive gameplay that resolves games faster.

5 Discussion

The experimental results reveal clear performance tiers among the agent architectures. Tool-Augmented agents achieve state-of-the-art performance with 90% win rate, significantly outperforming all other approaches. Parallel tool calling enables agents to synthesize recommendations, move validation, and game history in a single turn, providing comprehensive situational awareness.

Chain-of-Thought reasoning offers a compelling balance between performance (80% win rate) and transparency. Each decision is accompanied by an explicit reasoning chain that can be inspected and verified, addressing the interpretability concerns that plague black-box approaches.

The poor performance of In-Context Learning (no improvement over Baseline, highest error rate) suggests that static strategy guides cannot adapt to the dynamic nature of card game play. Retrieval overhead from scanning extensive guides may also disrupt decision timing.

However, game logs reveal a critical limitation: tool-augmented agents may over-rely on external tools rather than developing robust internal reasoning. Since symbolic tools are task-specific, this dependence may limit generalization to novel game states or environments. Future work should investigate hybrid architectures combining tool augmentation with explicit reasoning to achieve both high performance and strong generalization.

The 100% landlord win rate demonstrates that LLM agents can master asymmetric competitive roles when equipped with appropriate reasoning tools. These findings inform the design of LLM-based distributed systems where agents coordinate without central control, such as autonomous vehicle fleets or distributed sensor networks.

6 Conclusion

This work studied LLM agent architectures for multi-agent coordination in asymmetric card games. We compared four prompting strategies—Baseline, Chain-of-Thought, In-Context Learning, and Tool-Augmented—on Fight the Landlord (Dou Di Zhu), a 3-player asymmetric game requiring implicit coordination. Experiments ran 30 games per condition with rotating roles to ensure fair evaluation.

Results demonstrated that Tool-Augmented agents achieve state-of-the-art performance with 90% win rate, significantly outperforming Baseline (56.7%) and In-Context Learning (56.7%). Chain-of-Thought reasoning achieved 80% win rate with transparent, verifiable decision-making. Parallel tool calling enables agents to synthesize recommendations, move validation, and game history in a single turn. However, tool-augmented agents may over-rely on external tools, potentially limiting generalization. Chain-of-Thought reasoning offers a balance—producing transparent, verifiable reasoning while maintaining competitive performance.

The findings have broader implications for the design of LLM-based multi-agent systems. First, external tool augmentation can significantly enhance LLM performance in complex reasoning tasks, particularly those requiring precise rule adherence and multi-source information integration. Second, chain-of-thought reasoning remains a valuable approach for tasks requiring transparent, interpretable decision-making processes. Third, static in-context learning approaches may be insufficient for dynamic environments where adaptive strategy is crucial.

Future directions include hybrid architectures combining CoT reasoning with tool augmentation to achieve both high performance and generalization capability. Optimal tool combination

analysis could identify which tool configurations provide the greatest performance gains per computational cost. Larger-scale validation with 100+ games per condition would provide more robust statistical confidence in the results. Extension to other multi-agent environments such as negotiation systems, distributed sensor networks, and multi-robot coordination scenarios would establish the generalizability of these findings beyond the card game domain.

7 References

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